AMENORRHEA

Definition:

1- Primary amenorrhea:

Absent of menstruation by age of 14 year with failure of development of secondary sexual characters or absent menstruation by age of 16 year regardless secondary sexual characters.

2- Secondary amenorrhea: absent of menses for a length of time equivalent to a total at least 3 of previous cycle intervals, or more than 6 months duration.

We can classify amenorrhea: → Physiological or pathological / → Primary or secondary

A- <u>Physiological amenorrhea</u>:

1- Before puberty.
2- During pregnancy, pregnancy is the commonest cause of 2nd. amenorrhea during CBP.
3- During lactation.
4- After menopause.
5- Short periods of amenorrhea after menarche or before menopause.

B- <u>False (cryptomenorrhea)</u>: Menstruation occurs but escape of blood is prevented due to some obstruction ,like:

1- Cervical atresia :
   - Congenital.
   - Acquired following : Amputation , cauterization ,conization
2- Transverse vaginal septum.
3- The most common cause is imperforate hymen.

C: <u>True pathological types</u>:

I- <u>*Hypothalamic amenorrhea*</u>:

1- Pseudocyesis: حمل كاذب there is psychological upset, resulting in:

- amenorrhea.
- Abdominal distention
- Normal sized non-pregnant uterus.

It occurs in cases of intense desire or intense fear of pregnancy.

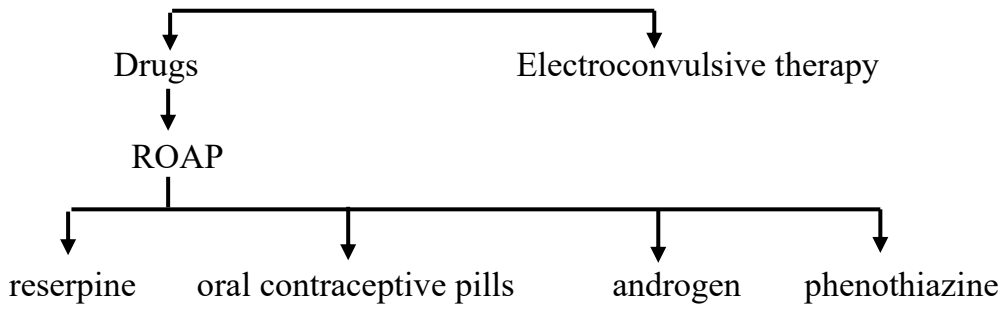
2- Anorexia nervosa: there is severe emotional disturbances.

resulting in:
- amenorrhea.
- loss of appetite.
- marked cachexia.

3- Bulimia: Psychological illness with disturbed body image occurs in young female 20 – 25 years. Female has polyphagia and then took purgative for fear of excessive weight gain.

4- Organic brain lesions: as infections, tumours, fracture base.

5- Iatrogenic amenorrhea induced by doctors



N.B.: Reserpine and phenothiazine cause amenorrhea through ↓ dopamine release in CNS. While OCP and androgen cause amenorrhea through ↓ FSH/LH.

N.B: Amenorrhea with OCP is due to :
- Continous intake of OCP.
- Post pill amenorrhea.

6- Excessive physical stress (Atheletic Amenorrhea).

7- Excessive mental stress

8- Chiari-frommel syndrome: amenorrhea + galactorrhea + small sized uterus after delivery.

9- Del-castillo syndrome: amenorrhea + galactorrhea + small sized uterus with out previous delivery.

10- Frohlich's syndrome: amenorrhea + genital hypoplasia + obesity + diabetes incipidus.

11- Laurence – Moon – Biedl's syndrome: in addition to feature of Frohlich's syndrome:

There are : - Polydactyly = additional finger.

- Syndactyly = 2 fingers attached together
- Retinitis pigmentosa. - Mental retardation.

12- Kalman's syndrome: Primary amenorrhea + Anosmia (Autosomal dominant disease).

II- *Pituitary amenorrhea*:

1- **Levi – lorian syndrome:** arrest of growth and hypogonadism due to decrease of growth hormone and gonadotropins.

2- **Simmond's and Sheehan's syndrome:**

Simmond's: Panhypopituitarism due to any cause in either sex., usually complete destruction.

Sheehan's: Panhypopituitarism due to severe hemorrhage especially post-partum hemorrhage → spasm and thrombosis of vessels of anterior pituitary so in female only, usually partial gland destruction.

Possible pathogenesis of Sheehan's (Why it occur due to obstetric not other similar causes):

A) Generally during pregnancy pituitary gland enlarges → vulnerable to ischemia.

B) During labour body needs oxytocin hormone (secreted from posterior pituitary) and so blood is shifted to the posterior lobe → anterior pituitary becomes relatively ischemic and very vulnerable to any shock → ischemia → necrosis.

3- **Tumours of anterior pituitary:**

a- Chromophobe adenoma:

- Secretory chromophobe adenoma, ↑ prolactin **(Forbes syndrome):**
  - Galactorrhea - Amenorrhea
  - Small sized uterus without previous pregnancy
- Non-specific non-secretory chromophobe adenoma which cause pressure atrophy on nearby cells.

b- Acidophil adenoma: Gigantism or acromegaly + Amenorrhea

c- Basophil adenoma:

- Cushing syndrome + Amenorrhea.
- Thyrotoxicosis + Amenorrhea.

III- *Ovarian Amenorrhea*

Which is due to either .

* hypofunction of ovary **or** * hyperfunciton of ovary

- Decrease estrogen and progesterone production
  - ↓
- Ovarian failure
  - Premature menopause
    - ↓
    - 1- Surgical removal
    - 2- Irradiation
    - 3- Chemotherapy
    - 4- Infection
    - 5- Autoimmune diseases
  - Streak ovary
    - ↓
    - Turner syndrome

- Continous production of estrogen with no cyclic pattern.
  - ↓
- Refractory ovary
  - ↓
  - 1- Follicular cysts and corpus luteum cyst.
  - 2- Feminizing tumour of ovary
  - 3- PCO.

IV- *Uterine amenorrhea*:

1- Congenital absence or marked uterine hypoplasia.

2- Surgical removal of uterus.

3- Intrauterine adhesion (synechia).

Asherman's Syndrome

1st described by Asherman's (1948) → amenorrhea traumatica atretica.

Causes :

1- Post abortive or post partum.
2- Over curettage.
3- T.B. endometritis.

Grades:

1- Corporeal
2- isthmocervical
3- Total

Types :

1- Mucus adhesion
2-Fibrous adhesion
3- Muscular adhesion.

C/P :

1- Amenorrhea and oligohypomenorrhea
2- Infertility.
3- Dysmenorrhea.

Investigation:

1- Hystrometry (sounding): we have multiple resistance.
2- HSG: → filling defects in uterine cavity.
3- Hystroscopy: visualization of adhesions in uterine cavity.

Treatment:

1- Cut the adhesion: by

a- Bevealed edge of sheath of hysteroscope.
b- Scissor. c- Resectoscope.

2- Avoid readhesion: by

a- Balloon of foly's catheter
b- Intrauterine device. c- Hydrocortisone.

3- Cyclic intake of estrogen and progesterone for 3 months to rebuild endometrium.

V- *General causes of amenorrhea (Intermediate organ factor):*

1- Prolonged debilitating diseases e.g. T.B.
2- Severe anemia.
3- Severe uncontrolled D.M.
4- Endocrine disorders e.g. hypo or hyper thyroidism/ Cushing or Addison's disease.
5- Marked obesity or marked loss of weight.
6- Liver cell failure or uremia.

N.B.: *Amenorrhea following childbirth:*

1- Chiari-frommel syndrome. 2-Sheehan's syndrome
3-Uterine adhesion (Asherman's syndrome). 4-General factor e.g. anemia

N.B.: Amenorrhea *galactorrhea syndromes:*

1- Chiaria – frommel syndrome 2-Del Castillo syndrome.
3-Forbes syndrome.

N.B.: *Primary Amenorrhea :*

I- Physiological: before puberty.

II- Pathological : (A) Cryptomenorrhea.

1- Congenital cervical atresia.
2- Transverse vaginal septum
3- Imperforate hymen.

B) True amenorrhea:

1- Hypothalamic cause: Pseudocyesis, anorexia nervosa, Frolich, Laurence Moon Biedl's syndrome.
2- Pituitary cause: levi lorian syndrome.
3- Ovarian cause: Turner's syndrome.
4- Uterine cause aplasia or hypoplasia. 5-Any of general cause

III- Pharmacological: Roap

N.B.: *Secondary amenorrhea:*

I- Physiological: during pregnancy, lactation after menopause.

II- Pathological: A- False amenorrhea: acquired cervical atresia

B- True amenorrhea:

1- Hypothalamic cause: pseudocyesis, anorexia nervosa, Chiari -Frommel and Del castillo.

2- Pituitary cause: Forbes syndrome.
Pituitary tumours.
Sheehan's syndrome

3- Ovarian. Cause: PCO
Premature menopause
Virilizing ovarian tumour.

4- Uterine cause: surgical removal of uterus, intra-uterine adhesions.

5- Any of general cause:

III- Pharmacological: Roap.

Investigation of a case of Amenorrhea:

I- History:

1- *Personal history:* *Age: 16-18 year: 1ry amenorrhea.
40 year: premature menopause.

* **Marital state*:*** as pregnancy is the commonest cause of 2nd amenorrhea.

2- *Menstrual history:* in 2ry type only.

3- *Obstetric history:* e.g. postpartum hemorrhage: Sheehan's syndrome peurpueral sepsis: Asherman's.,

4- *Past history:* of diseases
of operation
of drugs or irradiation

5- *Present history:* e.g.

- Type of amenorrhea.
- Cyclic lower abdominal pain, cryptomenorrhea.
- Galactorrhea.
- Hot flushes and sweating : menopause
- Hirsutism.

- Symptoms of pregnancy.

II- Examination

A- *General examination*:

1- Height : Short stature : levi-lorian, Turner's syndrome.

Tall : Gigantism.

2- Weight: Obese: Frolich's, laurance Moon Biedle's, PCO, Cushing's

Thin :Anorexia nervosa.

3- ABP:↑: Cushing's syndrome, PCO.

4- SSC: **5-Galactorrhea**. **6- Hirsutism**.

B- *Local exam*: - Inspect vulva for imperforate hymen.

- Bimanual exam: for uterus and adnexa.

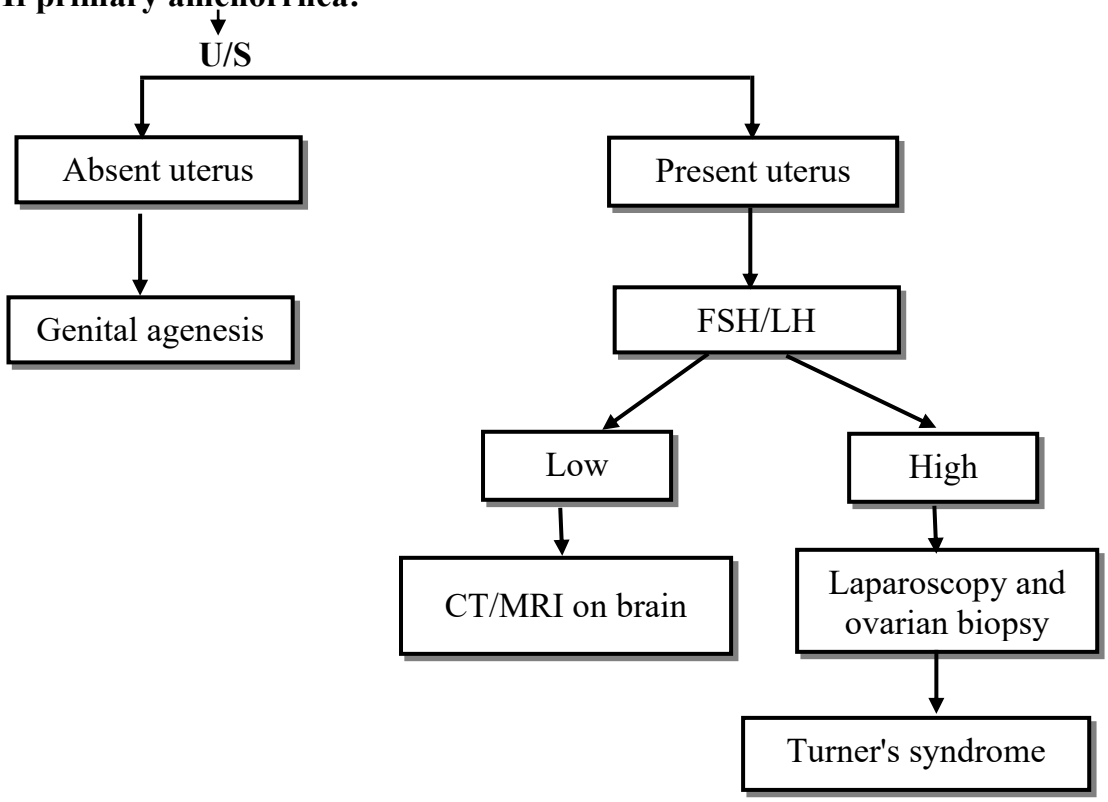
III- Protocol of investigation:

1- Exclude pregnancy by pregnancy test.

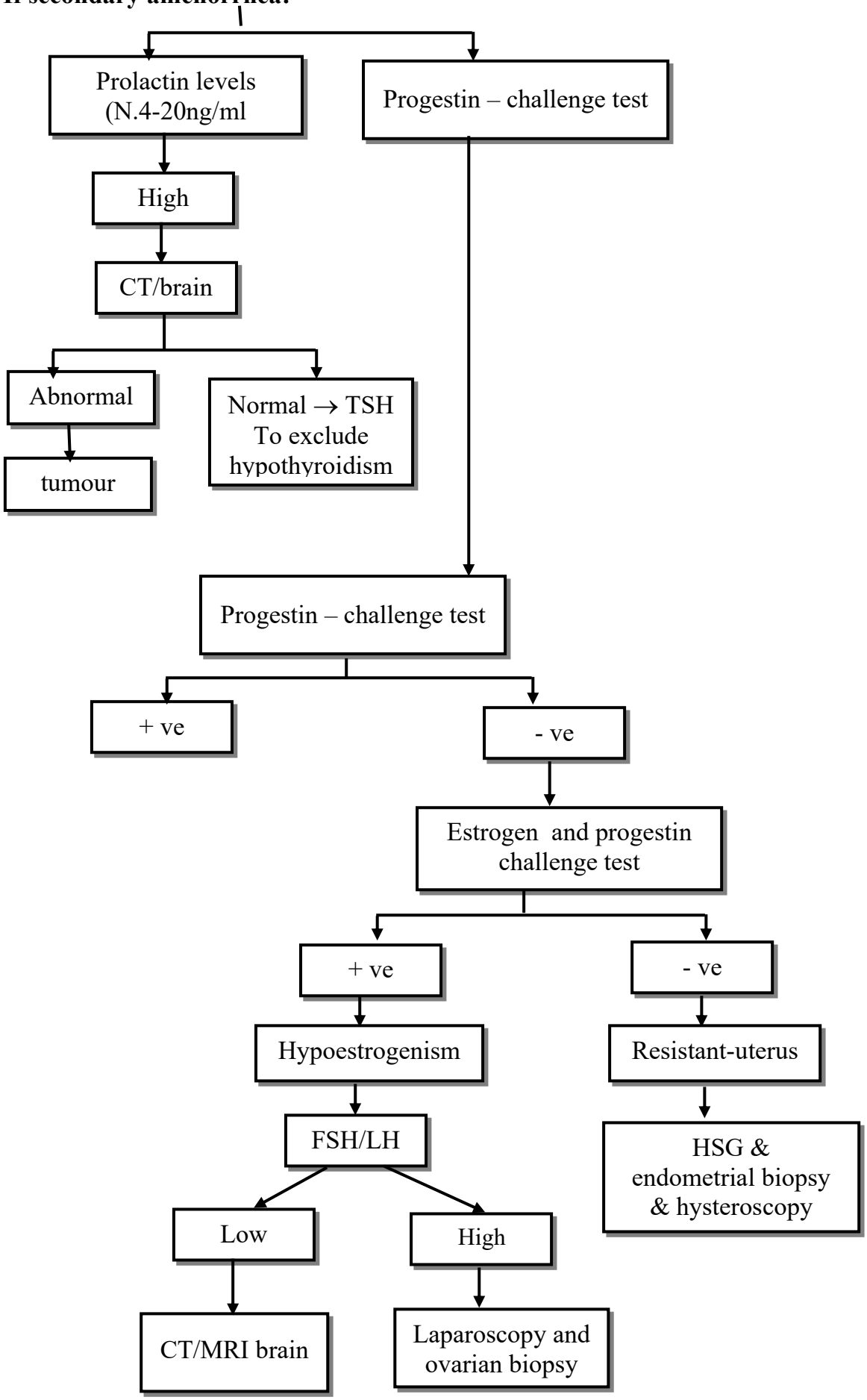
2- Exclude general cause of amenorrhea: CBC, blood glucose level.

3- Either primary or secondary amenorrhea.

If primary amenorrhea:

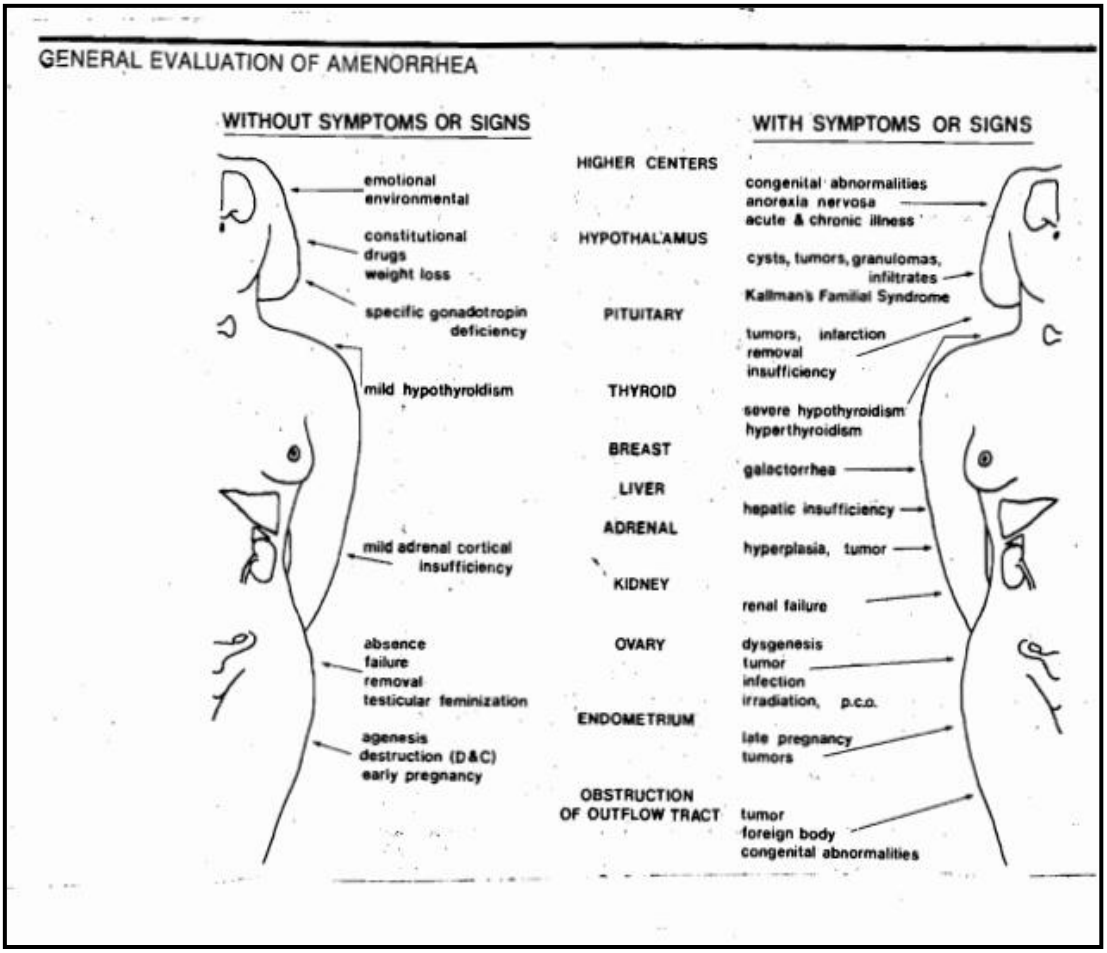


If secondary amenorrhea:



Treatment of Amenorrhea:

1- ttt of the cause (mention)

2- Amenorrhea due to hormonal dysfunction is treated by induction of ovulation (mention)

🗐 ***N.B.:*** Oligomenorrhea*:* abnormal ↑ length of cycle, leading to infrequent menses while hypomenorrhea: ↓ in amount and duration of bleeding. Causes of oligohypomenerrhea = causes of secondary amenorrhea.